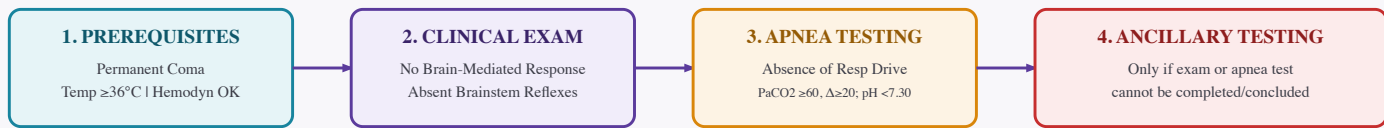


Brain Death Determination (BD/DNC)

AAN/AAP/CNS/SCCM 2023 Pediatric & Adult Consensus Guideline Reference.



1. Prerequisites & Stability

- **Etiology:** Catastrophic, permanent brain injury with a known cause and concordant imaging. Primary posterior fossa injury requires imaging evidence of catastrophic supratentorial injury.
- **Age/observation:** Do not evaluate below 37 weeks corrected gestational age. Under 24 months: wait ≥48 h after acute injury; age ≥24 months after hypoxic-ischemic injury: ≥24 h. Other injuries/interventions require sufficient observation for permanence.
- **Core Temp:** ≥ 36.0°C (96.8°F). Warm if hypothermic.
- **Hemodynamics:**
 - Adults: SBP ≥ 100 mmHg and MAP ≥ 75 mmHg (2023 AAN/AAP/CNS/SCCM)
 - Children: **SBP and MAP ≥5th percentile for age**; use the guideline age table. Individualize for a substantially different known chronic baseline.
- **Exclusions:** Exclude intoxication and CNS depressants using levels when available; if unavailable, allow ≥5 half-lives, longer with impaired clearance, obesity, or hypothermia. Pentobarbital must be <5 µg/mL or undetectable. Exclude paralysis with train-of-four or preserved deep tendon reflexes.
- **Metabolic:** Correct severe endocrine or electrolyte derangements.

2. Neurological Examination

- **Coma:** Complete absence of arousal. No motor responses to pain (spinal reflexes like triple flexion allowed).
- **Pupils:** No response to bright light in either eye; a fixed pupil diameter is not a criterion.
- **Corneal Reflex:** No blink to cotton swab contact.
- **Ocular reflex testing:** follow the complete current BD/DNC checklist. Oculocephalic testing requires cervical-spine and skull-base integrity; do not perform a head-turn maneuver when these are uncertain.
- **Oculovestibular testing:** use the full approved technique, including ear/skull-base safety, head position, irrigation duration and bilateral timing. Procedural mini-instructions are intentionally omitted here; consult the 2023 guideline supplemental checklist and correction.
- **Gag & Cough:** Absent gag (pharyngeal stim) and cough (tracheal suction catheter stim).
- **Facial Motor:** No grimace to TMJ or supraorbital pressure.
- **Infants <6 months:** No sucking or rooting reflex.

3. Apnea Testing Protocol

- **Preparation:** Correct hypoxemia, hypotension and hypovolemia. Preoxygenate with 100% O₂ ≥10 min to PaO₂ >200 mmHg; obtain baseline ABG. Without chronic hypercarbia, PaCO₂ 35–45 and pH 7.35–7.45.
- **Oxygenation:** Stop mandatory breaths and provide 100% O₂ by ventilator CPAP or a flow-inflating bag with functioning PEEP valve. Adult tracheal oxygen-insufflation technique differs; follow the full guideline and local checklist.
- **Chronic hypercarbia:** Start at the known chronic baseline PaCO₂. If suspected but baseline unknown, use the estimated baseline and require ancillary testing even if apnea criteria are reached.
- **ECMO:** Use the specialist protocol in recommendation 26, including cannula-specific ABG sampling. For VA ECMO, both patient and postoxygenerator PaCO₂/pH must meet criteria.
- **Observation/target:** No spontaneous breaths; ABG at 8–10 min. PaCO₂ ≥60 mmHg **AND** ≥20 above baseline (known chronic baseline if hypercarbic), and pH <7.30. Unmet targets require guideline-directed continuation/repeat or ancillary testing.

Abort: Any spontaneous breath means BD/DNC criteria are not met. Also stop for:

- Adult SBP <100 or MAP <75; child SBP or MAP <5th percentile despite support.
- Progressive oxygen-saturation decline below 85%.
- Arrhythmia with hemodynamic instability.

If instability makes abortion imminent, obtain an ABG before reconnecting when feasible; do not delay rescue.

4. Ancillary Testing Guidelines

Use for an incomplete/uninterpretable exam or apnea test, or specified situations such as uncorrectable metabolic confounders. Complete all assessable elements. **Do not bypass hypothermia, high sedative levels, or any brain-mediated activity with an ancillary test.**

ACCEPTED TESTS

- **Conventional 4-Vessel DSA:** Confirms absence of intracerebral blood flow.
- **Radionuclide Perfusion:** SPECT or planar radionuclide angiography using guideline-appropriate tracers.
- **TCD:** (Transcranial Doppler) **Adults only**; must show oscillating flow or spikes.

UNACCEPTABLE TESTS

- **EEG: NO LONGER ACCEPTED** (cannot evaluate brainstem).
- **CTA: NOT ACCEPTABLE** (insufficient validation).
- **MRI/MRA:** Not accepted.

BD/DNC DOCUMENTATION & REPEAT EXAM REQUIREMENTS

- ☒ **Adults:** Minimum one examination and one apnea test after the final examination; follow local policy and applicable law.
- ☒ **Pediatrics:** Two independent exams at least 12 hours apart, each followed by an apnea test (2023 guideline).
- ☒ **Rewarming Period:** If core temperature was ≤35.5°C, wait ≥24 hours after rewarming to ≥36°C before evaluation.
- ☒ **Time of Death:** When final apnea-test ABG results are reported meeting both PaCO₂ and pH criteria, with the clinical criteria met. If ancillary testing is required, use the time the attending documents a result consistent with BD/DNC.

Consensus Guideline: Greer DM, et al. Pediatric and Adult Brain Death/Death by Neurologic Criteria Consensus Guideline. *Neurology*. 2023;101(24):1112-1132. **PMID: 37821233** (<https://pubmed.ncbi.nlm.nih.gov/37821233/>).

AAN 2010 Guideline (prior; updated by 2023 BD/DNC consensus guideline): Wijdicks EF, et al. Evidence-based guideline update: determining brain death in adults. *Neurology*. 2010;74(23):1911-1918. **PMID: 20530327** (<https://pubmed.ncbi.nlm.nih.gov/20530327/>). (Found insufficient evidence on newer ancillary tests; the Section 4 accept/reject criteria follow the 2023 consensus guideline.)